

**EXP.NO: 5**

**Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.**

### **AIM:**

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

### **ALGORITHM:**

#### **1. Input:**

- The user fills out a web form with their **name** and **email**.
  - The form will use both **GET** and **POST** methods for data submission.
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#### **2. Steps for Creating and Handling the HTTP GET and POST Requests:**

##### **1. Create an HTML Form:**

- Design an HTML form with two input fields: one for the **name** and one for the **email**.
- Create two buttons to allow the user to submit the form using **GET** or **POST** methods.
- Use the action attribute to define the **Servlet URL** and specify the form method (either GET or POST).

##### **2. Create the Servlet to Handle GET and POST Requests:**

- Write a Java Servlet class that will handle **both GET and POST requests**.
- The servlet will have two methods:
  - doGet() to handle the GET requests.
  - doPost() to handle the POST requests.
- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

### 3. Handle GET Requests in the Servlet:

- In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
- Output the data as a response, including the **name** and **email** in the URL (visible to the user).

### 4. Handle POST Requests in the Servlet:

- In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
- Output the data as a response, displaying the **name** and **email** without exposing them in the URL (more secure).

### 5. Return the Output:

- The Servlet should return an HTML page displaying the **name** and **email** entered by the user.
- For the **GET method**, the output should be visible in the URL.
- For the **POST method**, the output should be visible in the response body, not the URL.

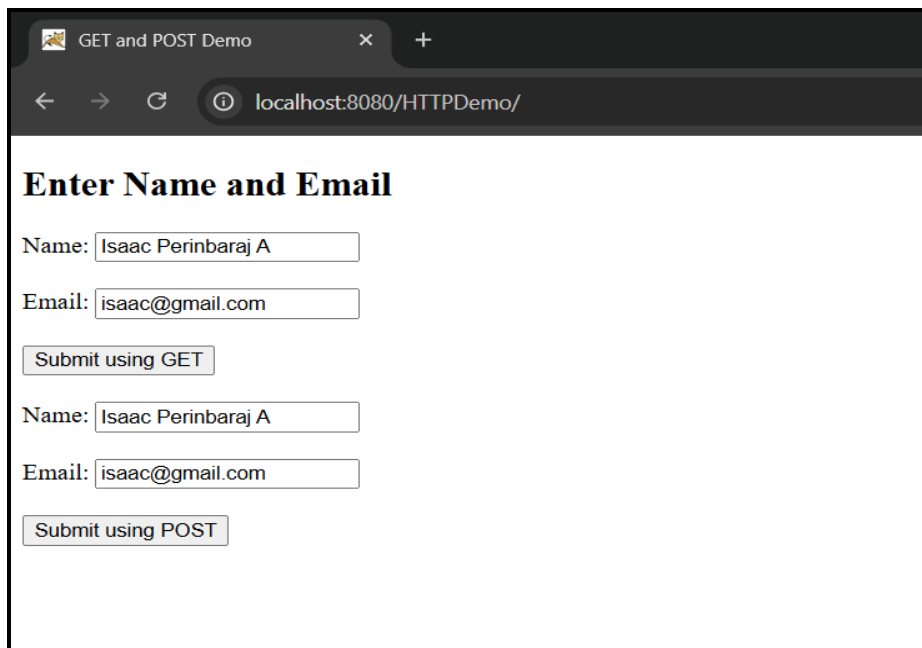
### 6. Deploy and Test the Servlet:

- Compile the Servlet class and deploy it to a **Servlet container** (like **Apache Tomcat**).
- Test the form using **GET** and **POST** methods to see the difference in data submission.

### 7. Observe and Compare GET and POST Results:

- Test the **GET method**: The form data is sent in the URL.
- Test the **POST method**: The form data is sent in the HTTP request body, not visible in the URL.

## Output:



GET and POST Demo

localhost:8080/HTTPDemo/

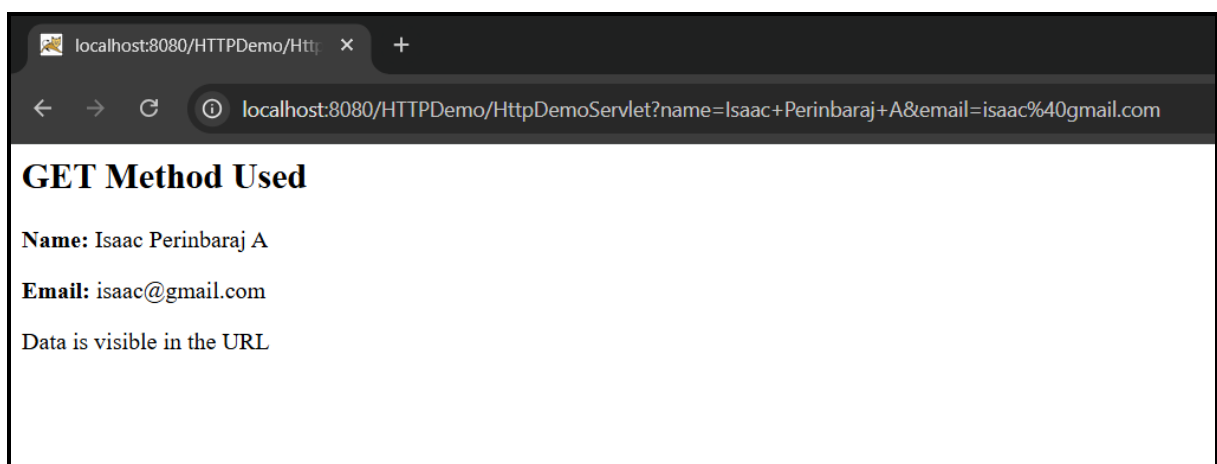
### Enter Name and Email

Name:

Email:

Name:

Email:



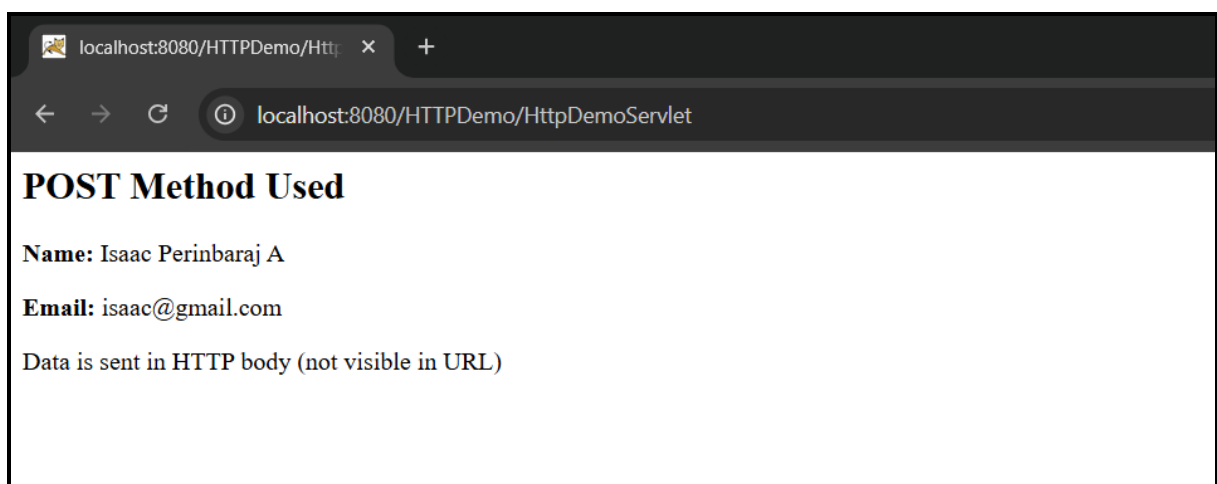
localhost:8080/HTTPDemo/HttpDemoServlet?name=Isaac+Perinbaraj+A&email=isaac%40gmail.com

### GET Method Used

**Name:** Isaac Perinbaraj A

**Email:** isaac@gmail.com

Data is visible in the URL



localhost:8080/HTTPDemo/HttpDemoServlet

### POST Method Used

**Name:** Isaac Perinbaraj A

**Email:** isaac@gmail.com

Data is sent in HTTP body (not visible in URL)

## HTML File (index.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>GET and POST Demo</title>
</head>
<body>
  <h2>Enter Name and Email</h2>
  <form action="HttpDemoServlet" method="get">
    Name: <input type="text" name="name"><br><br>
    Email: <input type="text" name="email"><br><br>
    <input type="submit" value="Submit using GET">
  </form>
  <br>
  <form action="HttpDemoServlet" method="post">
    Name: <input type="text" name="name"><br><br>
    Email: <input type="text" name="email"><br><br>
    <input type="submit" value="Submit using POST">
  </form>
</body>
</html>
```

## Java File (HttpDemoServlet.java):

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class HttpDemoServlet extends HttpServlet {

    // Handle GET request
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");

        out.println("<html><body>");
        out.println("<h2>GET Method Used</h2>");
        out.println("<p><b>Name:</b> " + name + "</p>");
        out.println("<p><b>Email:</b> " + email + "</p>");
        out.println("<p>Data is visible in the URL</p>");
        out.println("</body></html>");
    }
}
```

```

// Handle POST request
protected void doPost(HttpServletRequest request, HttpServletResponse
response)
    throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    String name = request.getParameter("name");
    String email = request.getParameter("email");

    out.println("<html><body>");
    out.println("<h2>POST Method Used</h2>");
    out.println("<p><b>Name:</b> " + name + "</p>");
    out.println("<p><b>Email:</b> " + email + "</p>");
    out.println("<p>Data is sent in HTTP body (not visible in URL)</p>");
    out.println("</body></html>");
}
}

```

#### File (web.xml):

```

<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
    version="5.0">

    <servlet>
        <servlet-name>HttpDemoServlet</servlet-name>
        <servlet-class>HttpDemoServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>HttpDemoServlet</servlet-name>
        <url-pattern>/HttpDemoServlet</url-pattern>
    </servlet-mapping>

</web-app>

```

#### RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly